

BUS311 Lemonade Stand Finance Model

Prepare - Model - Analyze - Decide

Reserved for later use — not assigned in M04

No submission is required until your instructor assigns this activity.

From a simple stand to an integrated finance decision

Use this guide with the M04 Lemonade Stand workbook and Bill Ackman's finance lesson:

<https://www.youtube.com/watch?v=WEDlj9JBTC8>

What you will learn

- Translate financing choices into cash, debt, equity, and ownership.
- Connect revenue, expenses, profit, cash flow, and the balance sheet.
- Explain why net income and operating cash flow can move differently.
- Evaluate growth, risk, valuation, and dilution with correctly labeled metrics.

Your deliverable

When assigned, enter assumptions and trace the supplied formulas. This is model interpretation, not a formula-building assessment. Complete prompts 1–4 for the operations/cash-flow stage; complete valuation and prompt 5 after those concepts have been taught.

Staged route

Prepare before the application session

- Watch the video and capture its base assumptions before class.
- Enter base assumptions and the fixed stand schedule. The later downside-growth input is separate.

Stage A: statements, margins, and cash

- Trace Day 1 funding, Day 2 investment, and Year 1 operations.
- Explain gross profit versus net income and profit versus cash.
- Discuss the two explicit accounting conventions below.
- Complete interpretation prompts 1, 3, and 4 when assigned.

Stage B: operating sensitivity

- Enter a lower unit-growth assumption and compare Year-5 revenue and EBIT.
- Explain why fixed labor and depreciation make profit sensitive to sales.
- Complete interpretation prompt 2.
- The stand schedule remains fixed. This test neither ranks every growth driver nor establishes funding feasibility.

Later: valuation and ownership

- Return to EPS, P/E, equity value, and share transactions after these topics are taught.
- Complete interpretation prompt 5.

The instructor will specify timing and submission requirements. Do not combine this full activity with the M04 ratio assignment.

Build the statements in a connected order

Funding

Investor shares multiplied by price per share produce investor cash. Debt adds cash but also creates interest expense and a senior claim.

Asset transformation

Buying a stand and inventory changes the mix of assets. It does not change total assets by itself.

Operating result

Revenue = cups sold multiplied by price per cup.

Gross profit = revenue minus cost of goods sold.

Operating profit before depreciation = gross profit minus labor.

EBIT = operating profit before depreciation minus depreciation.

EBT = EBIT minus interest expense.

Cash bridge

Start with net income, add back noncash depreciation, and include the release of opening inventory. This is why cash can increase even when Year-1 net income is negative.

Two accounting judgments are intentionally visible

Year-1 tax benefit

The video-replication case rounds the tax effect of the Year-1 loss and treats the benefit as immediately available cash. That produces the video's ending-cash result.

The conservative comparison records no immediate tax benefit. Recognition depends on whether the business can support future taxable income or obtain a current refund.

Do not hide this difference. Explain the assumption behind the result you use.

Founder idea and illustrative goodwill

The video assigns value to the founder's idea and labels it goodwill so the opening balance-sheet equation is easy to see.

Internally generated goodwill ordinarily is not recorded this way under standard financial reporting. Treat it as an explicit teaching convention, not a general accounting rule.

Expansion schedule

The workbook follows the video's fixed schedule of 1, 2, 3, 4, and 7 active stands. It does not automatically buy a stand whenever cash reaches a particular threshold.

Evaluate performance with the right labels

Profitability

- Operating margin = EBIT divided by revenue.
- Net margin = net income divided by revenue.
- Earnings yield = EPS divided by share price.

Investment efficiency

The activity reports a video-style return on gross stand cost: Year-5 EBIT divided by cumulative stand purchases.

This is a useful teaching comparison, but it is not a complete conventional ROIC calculation. A full ROIC analysis would define NOPAT and invested capital consistently.

Growth

Do not calculate a conventional percentage growth rate from a negative earnings base. Mark the comparison N/M until a positive, meaningful base exists.

Historical rate context

The video's 3% risk-free comparison reflects the lesson's historical context. It is not a current market-rate assumption.

Valuation and share transactions

Earnings to value

EPS multiplied by a selected P/E multiple produces an illustrative share price.

Share price multiplied by shares outstanding produces illustrative equity value.

Primary issuance

When the company issues new shares, company cash and shares outstanding both increase. Existing owners are diluted because the ownership denominator becomes larger.

Founder secondary sale

When the founder sells existing personal shares, the founder receives the cash. Company cash and total shares outstanding do not change.

These transactions can involve the same number of shares and price but have different economic consequences. Always identify who receives the cash.

Final submission check

- Every base input is populated; downside growth is entered when that stage is assigned.
- All statement and ownership checks pass; PASS is an internal-consistency check, not a completion grade.
- The sensitivity case recalculates from its own growth assumption.
- The interpretation responses for the assigned stage are complete.
- No formula cell has been overwritten.

Source transcript used for cross-checking the case assumptions:

<https://singjupost.com/everything-need-know-finance-investing-william-ackman-transcript/>